Software Requirements Specification (SRS) - IEEE 830 Standard

SoluoPrint - Premium All-in-One Print Shop Management (Print/Job management, CRM, Financial Management & Analytics Platform, Automated SMS Notifications)

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the SoluoPrint platform. SoluoPrint is a web-based, multi-tenant print shop ERP engineered to automate pricing calculations, production tracking, customer relations management (CRM), and financial accounting for specialized digital studios.

1.2 Scope

The system scope covers single and bulk job creation, custom dimension calculators (supporting feet, inches, cm, mm, and meters), real-time production status tracking, a self-service customer intake portal, email and SMS notification engines (smsonlineapi / SMTP), and integrations with digital payment gateways (Hubtel, Paystack, Flutterwave).

1.3 Definitions, Acronyms, and Abbreviations

• ERP: Enterprise Resource Planning.

• MoMo: Mobile Money.

• RLS: Row Level Security.

• SRS: Software Requirements Specification.

• SMSGH: Telephony messaging protocol.

1.4 References

IEEE Computer Society. (1998). IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications. IEEE.

Supabase Documentation. (2026). Row Level Security & Storage. Supabase Inc.

2. Overall Description

2.1 Product Perspective

SoluoPrint is a standalone multi-tenant web application that replaces spreadsheets and generic accounting systems. It integrates with external SMS services (smsonlineapi) and payment aggregators (Paystack) to deliver a unified workspace.

2.2 Product Functions

Key functions include automated pricing, job workflow tracking (Pending, In Progress, Completed, Delivered), customer balance management, ledger expense/revenue logging, and multi-format analytics reporting.

2.3 User Characteristics

• Shop Owner: Requires high-level financial health dashboards and settings control.
• Press Operator: Requires simplified job lists and production tracking.
• Customer: Requires self-service upload tools and payment checkout.

2.4 Constraints

The application requires internet connectivity to communicate with the Supabase database. Upload file sizes are constrained up to 500MB via Namecheap cPanel PHP configuration.

3. Specific Requirements

3.1 Functional Requirements

FR-01: Dual-Provider Admin Authentication using Supabase Auth with RLS policies.
FR-02: Customer Portal Dual Auth using custom customer credentials.

FR-03: Single and Bulk Job Intake with automatic square footage calculation.
FR-04: Preset Size Configuration to automate width/height fields in forms.
FR-05: Real-time Production tracking across production phases.
FR-06: Telephony Integration using smsonlineapi for transactional notifications.
FR-07: Payment Clearing & Ledger recording against outstanding balances.
FR-08: In-App notification system triggered on job creations or portal uploads.
FR-09: Financial report compilation (PDF/Excel/PNG export) for P&L, Revenue, and Expenses.
FR-10: Activity Logging & Audit Trail to monitor user logins and operations.

3.2 Non-Functional Requirements

• NFR-01: Performance & Latency: Dashboard metrics and financial aggregations must load within 1.5 seconds under a load of 100 concurrent requests.
• NFR-02: Security: All persistent tables must enforce Supabase Row Level Security (RLS) policies. Sensitive database credentials and API keys must be isolated in cPanel environment variables.
• NFR-03: Reliability & Fault Tolerance: The frontend application must include a ConnectivityManager to handle network drops and reconnect to database subscriptions seamlessly.
• NFR-04: Portability: The system must render consistently across standard modern web browsers (Chrome, Safari, Firefox, Edge) using a responsive CSS layout.

4. System Architecture & Diagrams

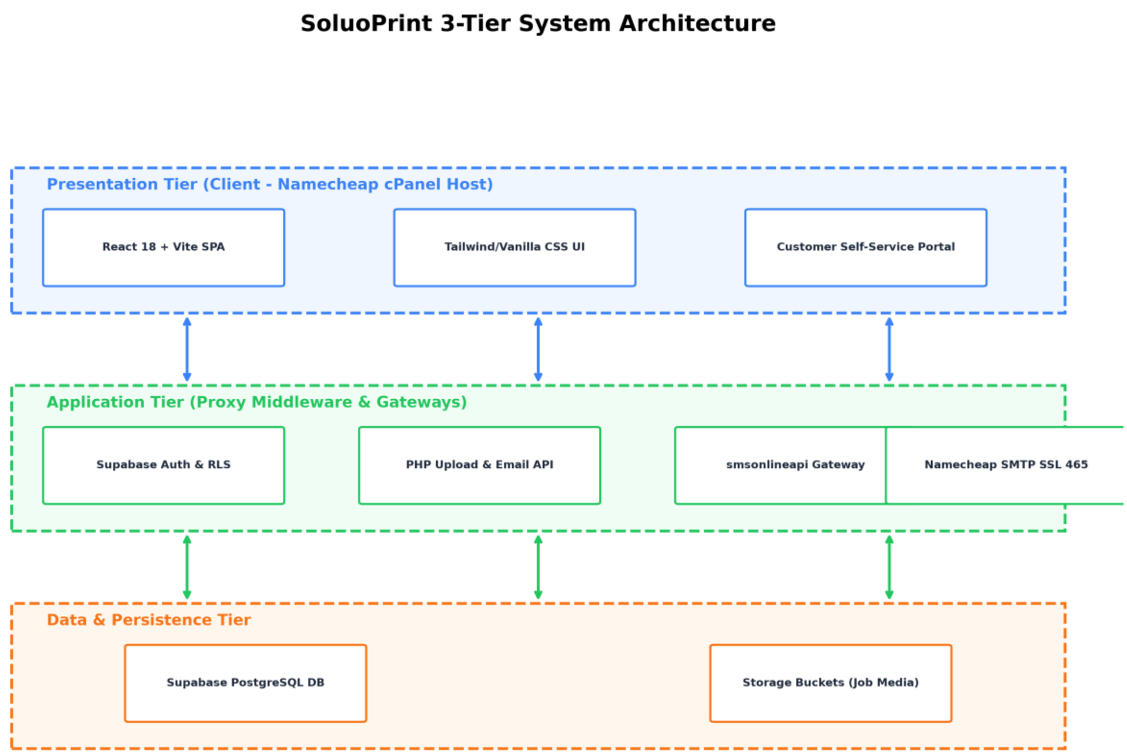

Figure Evidence: 3-Tier System Architecture Diagram

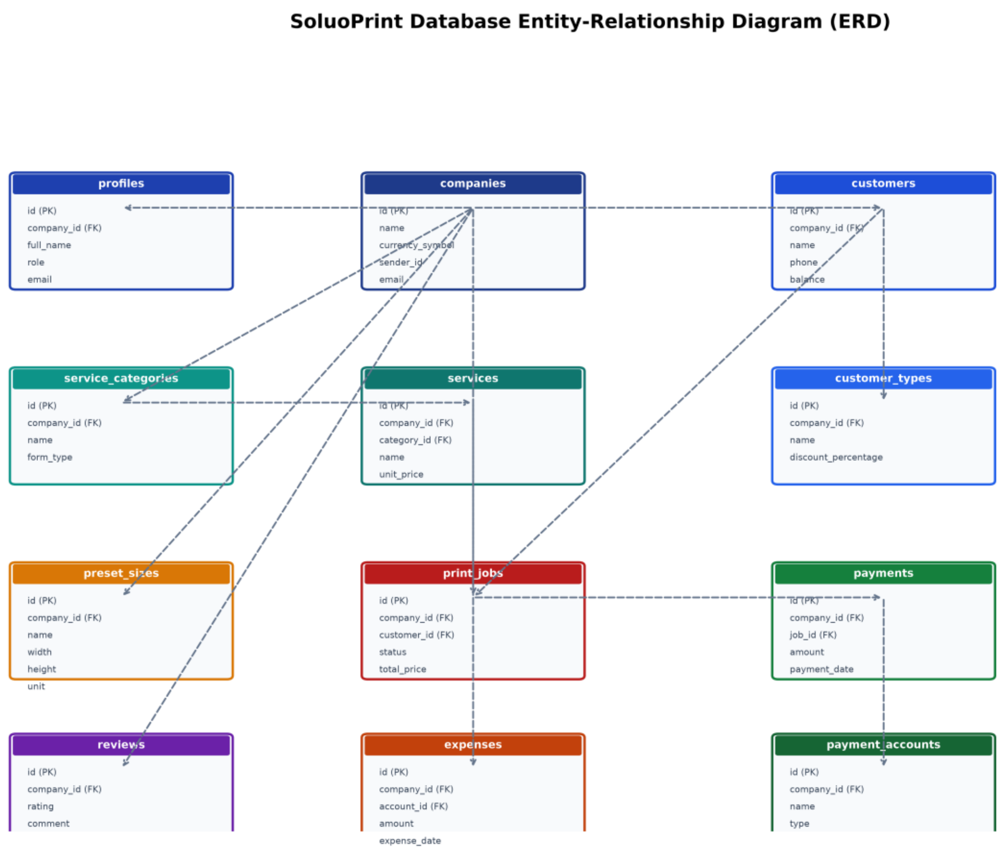

Figure Evidence: Normalized Entity-Relationship Diagram (ERD)

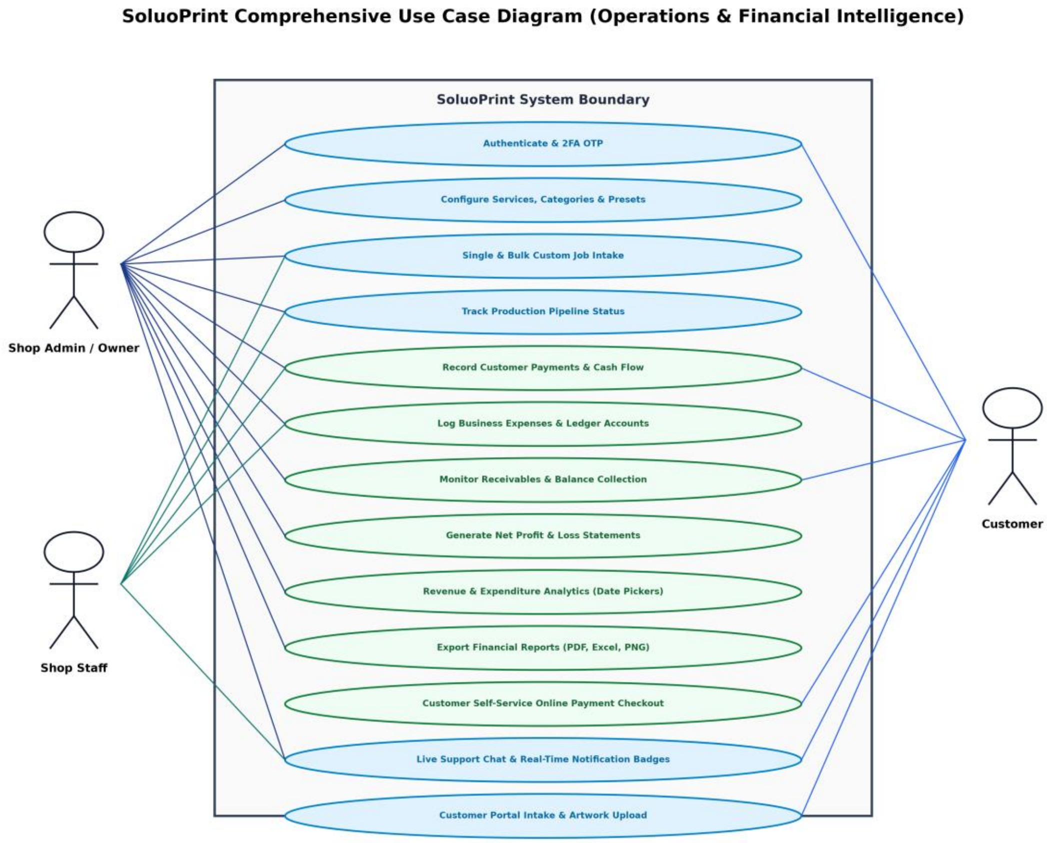

Figure Evidence: System Use Case Diagram